Data quality: what the validation rules caught

37,538 NZ interruption events checked against 6 rules, 870 exceptions logged before any reliability calculation

37538
Total Events

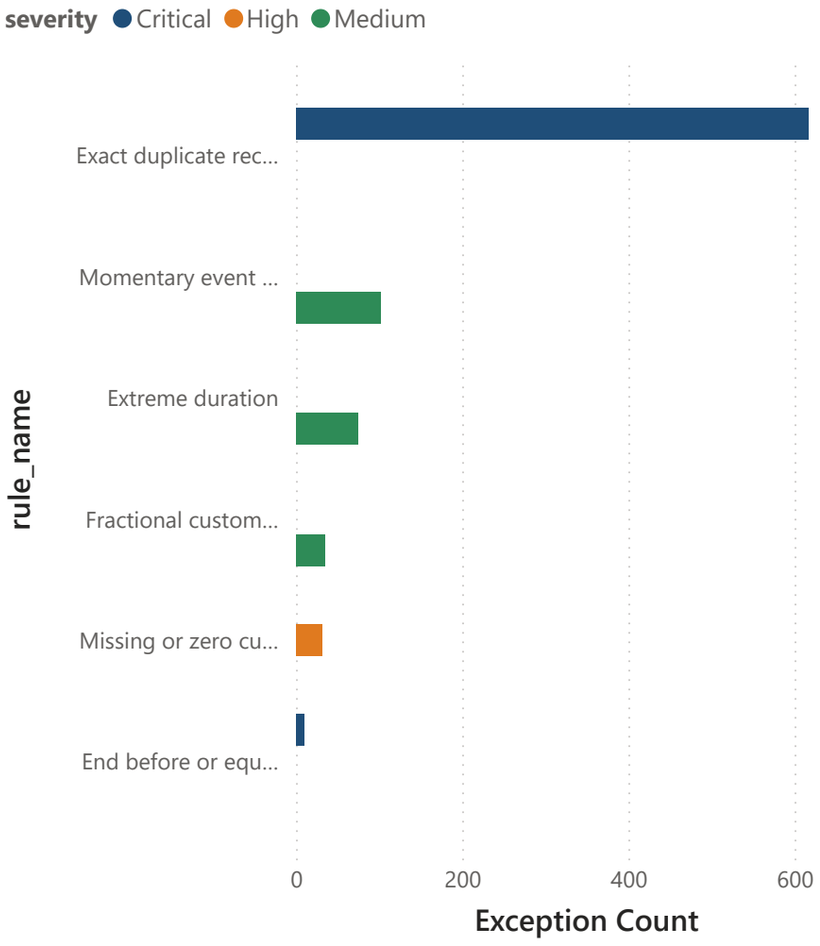
870
Exception Count

868
Events With Exceptions

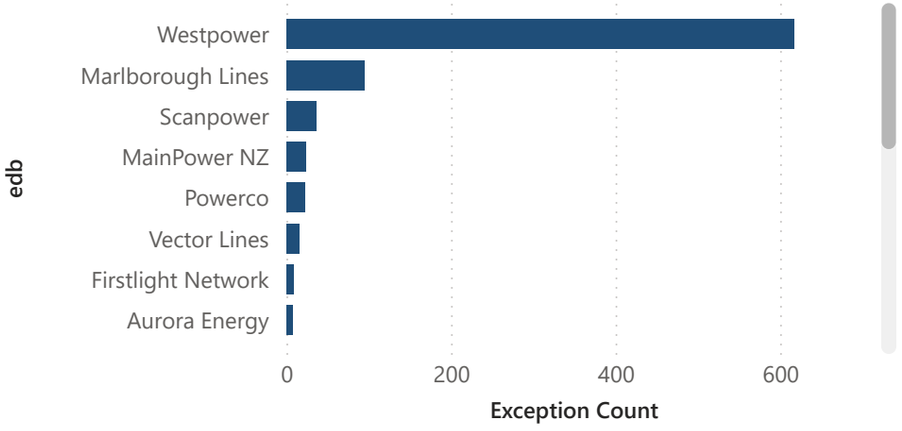
2.3%
Exception Rate

rule_id
- [ ] V01
- [ ] V02
- [ ] V03

Exceptions by rule



Exceptions by distributor



Findings

870 exceptions across 868 events. Duplicates (V07) are the largest group at 617, all Westpower.

Westpower's duplicate rows double its SAIDI from 184.6 to 369.1. The duplicates are in the published source, not our load.

7 distributors report customer minutes per customer, not per ICP as the dictionary says. A naive 5% rule would raise about 12,900 alerts, about 99% of them false.

V02 (day/month swap) and V06 (customer minutes mismatch) are designed but not applied, listed in the rules catalogue.

Reliability: Powercor SAIDI and Major Event Days

Powercor daily SAIDI, FY2020 to FY2024. MED threshold rebuilt from published data (6.759 min), 16 of 16 MED days match Powercor's own flags

16
MED Days

122.99
CAIDI

Findings

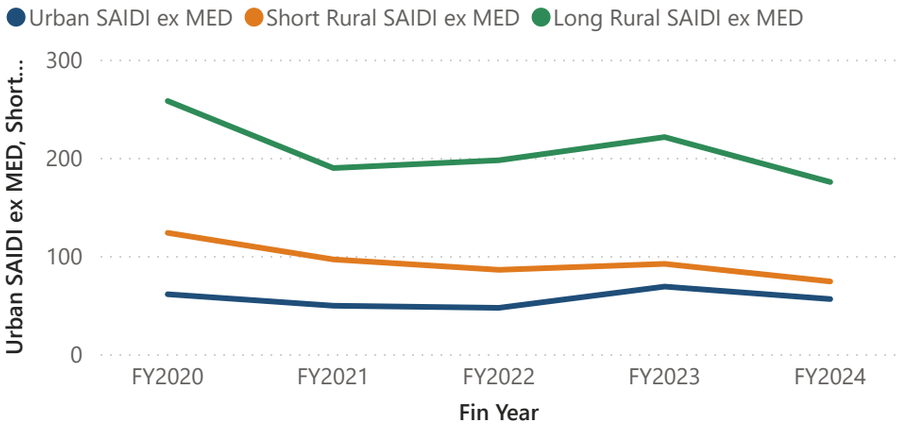
MED threshold rebuilt from Powercor's daily data: 6.759 min. 16 of 16 MED days match, 0 mismatches.

Exclusions are applied first, then the day is tested. 31 Jan 2020: 6.98 raw, 3.95 after exclusions, so not a MED.

STPIS targets reproduced exactly (5 year average ex MED): Urban 56.378, Short Rural 94.273, Long Rural 208.154.

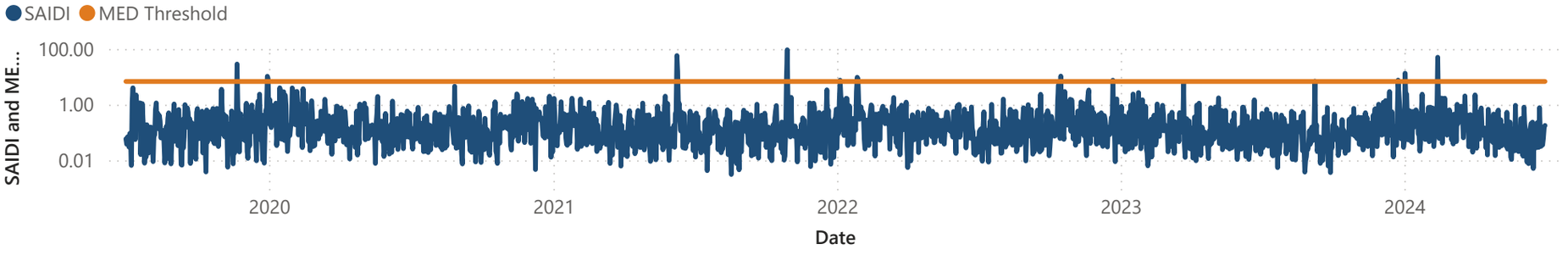
Storm days drive rural reliability: Long Rural FY2022 was 481.8 min with MEDs vs 197.4 without.

SAIDI ex MED by feeder class



Fin Year	MED Days	SAIDI	SAIDI ex MED	CAIDI	Urban SAIDI ex M
FY2020	*3*	*179.88*	*132.87*	*114.02*	*60.*
FY2021	*2*	*174.14*	*101.10*	*131.16*	*49.*
FY2022	*5*	*225.78*	*98.58*	*136.68*	*47.*
FY2023	*2*	*132.56*	*114.54*	*106.22*	*68.*
FY2024	*4*	*169.25*	*91.37*	*124.18*	*56.*
Total	***16***	***881.61***	***538.46***	***122.99***	***281.***

Daily SAIDI vs MED threshold (log scale)



GSL Reconciliation

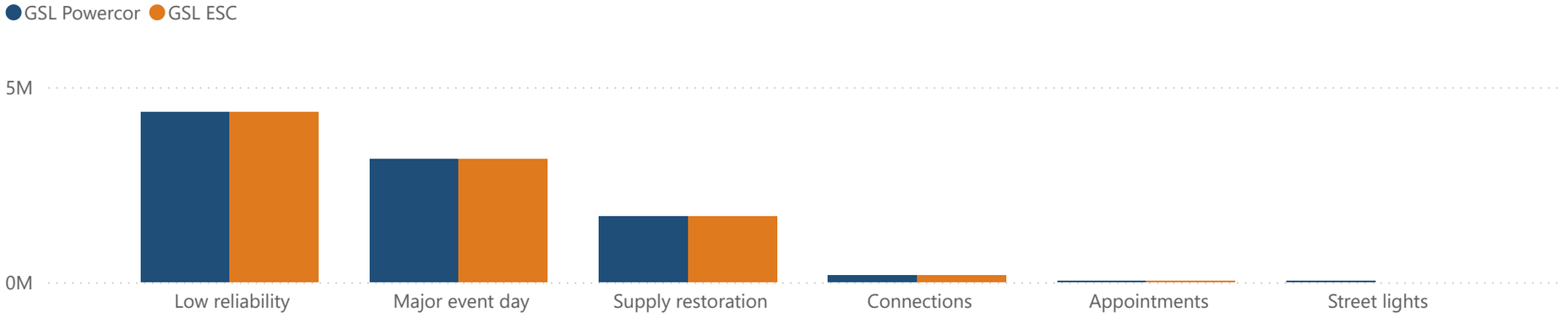
Powercor's GSL model submitted to the AER (MOD 9.04) compared with the ESC's published GSL figures, FY22 and FY23 actuals

10
Rows Matched

12
Rows Total

GSL payments by category: Powercor AER model vs ESC published

FY22 and FY23 actuals. Equal bars mean a match. Street lights is not in the ESC table (scope difference, not an error).



Financial year

- ☐ FY22
- ☐ FY23

Data sources

Powercor: PAL MOD 9.04 GSL model, submitted to the AER (Jan 2025).

ESC Victoria: GSL annual performance data for distributors.

Logic lives in SQL view rpt.gsl_reconciliation (FULL OUTER JOIN on a shared category map).

Year	Category	Powercor count	ESC count	Powercor $	ESC $	Difference $	Status
FY22	Appointments	310	310	10,850.00	10,850.00	0.00	Match
FY22	Connections	301	301	78,240.00	78,240.00	0.00	Match
FY22	Low reliability	17,232	17,232	2,320,950.00	2,320,950.00	0.00	Match
FY22	Major event day	27,479	27,479	2,473,110.00	2,473,110.00	0.00	Match
FY22	Street lights	30		750.00		750.00	Only in Powercor
FY22	Supply restoration	4,961	4,961	706,370.00	706,370.00	0.00	Match
FY23	Appointments	179	179	6,265.00	6,265.00	0.00	Match
FY23	Connections	306	306	87,600.00	87,600.00	0.00	Match
FY23	Low reliability	14,738	14,738	2,029,520.00	2,029,520.00	0.00	Match
FY23	Major event day	7,579	7,579	682,110.00	682,110.00	0.00	Match
FY23	Street lights	23		575.00		575.00	Only in Powercor
FY23	Supply restoration	6,994	6,994	973,810.00	973,810.00	0.00	Match
Total		**80,132**	**80,079**	**9,370,150.00**	**9,368,825.00**	**1,325.00**	

Findings

10 of 10 comparable rows match to the dollar.

Street lights appear only in Powercor's model. The ESC table does not publish street light GSLs, so this is a scope difference, not an error.

Inside Powercor's model, rate x count = amount on every actual row (0 exceptions).

Tolerance: $1 and 0.5 count, to allow for rounding.